GGG-K-494B
April 10, 1974
SUPERSEDING
Int. Fed. Spec. GGG-K-00494a
January 11, 1968 and
Fed. Spec. GGG-K-494
September 1, 1953

FEDERAL SPECIFICATION

KNIVES; GENERAL PURPOSE SHOP, DETACHABLE BLADE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers general purpose shop knives with detachable blades which are used for cutting cardboard, gaskets, upholstery, rubber, leather, and similar non-metallic materials.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification. Knives shall be as follows and as specified (see 6.1):

General Purpose Shop Knife

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-P-40 - Packaging and Packing of Hand Tools.
QQ-S-700 - Steel Sheet And Strip, Medium And High Carbon.

Federal Standard:

Fed. Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook:

H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of knives which are otherwise in accordance with this specification.

3.2 Material. Material shall be as specified herein. Material not definitely specified shall be of a good commercial quality, ordinarily used for the purpose. Material shall be free from all defects and imperfections.

3.3 Marking. Each knife shall be marked in a plain and permanent manner with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.

3.4 Handles. Handles shall be well constructed, well balanced, and provide a comfortable handgrip. Handles shall be free from burrs, sharp edges, or other defects which might cause injury or discomfort to the user. A safe and easily accessible storage compartment for five blades shall be provided in the handle. Handle design shall provide the user adequate protection for both safely retaining and positioning the cutting blade during usage of the knife. The handle shall comply with tests in 4.5.1.

3.5 Blade adjustment and alignment guide. The blade alignment guide shall be designed to slide in and out of the handle freely (without binding) and to position and hold the blade securely in place. A positive blade stop shall be incorporated into the blade alignment guide mechanism so that the blade can be locked securely in at least three or more separate blade positions. One position shall lock the blade in the fully extended position with at least one inch of the blade protruding out of the handle (see figure 1); another position shall lock the blade fully within the handle (none of the blade exposed); and the other position or positions shall lock anywhere between these two extremes.

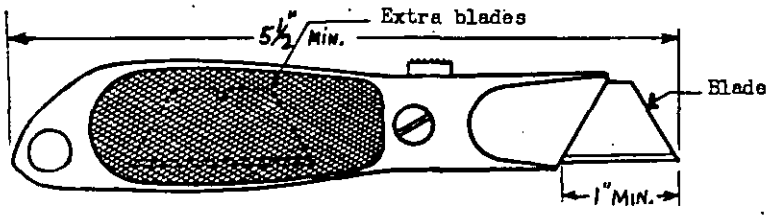


FIGURE 1. General purpose shop knife.

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3.6 Blades. The blades shall be of high carbon steel conforming to QQ-S-700 steel number SAE 1095. The blades shall be hardened, tempered, and ground. The Rockwell hardness of the blades shall be not less than the equivalent of 56 nor more than 61 on the C scale when tested as specified in 4.5.2. The blades shall be double ended, and provided with a single cutting edge. Five blades shall be furnished with each knife. Blades shall be not less than 0.025 inch thick and 3/4 inch wide by 1-7/8 inches long (at the cutting edge).

3.7 Finish.

3.7.1 Handles. Handles shall be plated or painted in accordance with manufacturer's standard practice. All plated or painted surfaces shall be free from unplated spots, unpainted spots, blisters, or other defects which might affect the serviceability of the knives.

3.7.2 Blades. Blades shall be provided with a protective petroleum coating to prevent rusting.

3.8 Screw threads. All threads shall conform to H28 as to shape and form of threads.

3.9 General purpose shop knife.

3.9.1 General purpose shop knives shall be in accordance with the dimensions of, and similar to figure 1, with blade in fully extended position. Knives shall be provided with a push button or other quick means of manual adjustment for extending and locking the cutting blade and retracting the blade completely into the handle without disassembly of the handle or the use of tools.

3.10 Sheath (holster). When specified (see 6.1), a sheath (holster) shall be furnished with each knife. The sheath shall be designed for carrying the knife on the user's belt. The sheath material shall be suitable for the purpose intended, and be formulated from either leather or plastic at the option of the manufacturer.

3.11 Workmanship. The blade alignment guide shall slide in and out of the handle freely (without binding) and blade shall lock securely in the three separate positions described in 3.5. Knives shall be free from rust, burrs, fins, and any imperfections which may impair their appearance, serviceability, or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for lot acceptance.

4.2.1 Lot. Consider all knives of the same type submitted for acceptance at one time as a lot for purposes of inspection.

4.2.2 Sampling for examination. Select a random sample from each lot for examination in accordance with MIL-STD-105 at inspection level II. The acceptance quality level (AQL) shall be 4.0 percent defective.

4.2.3 Sampling for acceptance tests. Select a random sample from each lot for acceptance tests in accordance with MIL-STD-105 at inspection level S-3. The AQL shall be 2.5 percent defective.

4.3 Inspection. Select each of the sample knives in accordance with 4.2.2 and visually and dimensionally examine to determine compliance with this specification. Reject any knife in the sample containing one or more defects and if the number of defective knives in any sample exceeds the acceptance number for the sample, reject the lot represented by that sample.

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4.4 Acceptance tests. Select each sample knife in accordance with 4.2.3 and test in accordance with 4.5 to verify compliance with this specification. Reject any knife in the sample which does not meet the requirements for any of these tests and if the number of nonconforming knives exceeds the acceptance number for that sample, reject the lot represented by the sample.

4.5 Tests.

4.5.1 Handle drop test. The handle shall not break, crack, or otherwise become defective when dropped on a concrete floor from a height of not less than 8 feet above the floor.

4.5.2 Hardness. The blade of each of the knives selected in accordance with 4.2.3 shall be tested in accordance with Method 243 of Fed. Std. No. 151 to determine compliance with 3.6. Test as close as practicable to the cutting edge.

4.5.3 Blade and handle pressure test. Holding the knife handle at a 45° angle (see figure 2) and the blade locked in the fully extended position, apply 50 lbs. of downward pressure on the tip of the blade for at least 5 seconds. During this test the (2) halves of the handle (at the blade end) shall not show any visible signs of spreading apart; nor the blade break or show any signs of either slipping or pivoting out of the locked position.

4.5.4 Cutting test. Knife blades shall be subjected to a cutting test on hard woods such as knots found in seasoned black spruce or yellow pine. Sufficient cuts shall be made with the knife blades to indicate that the cutting edge will not chip, dull, or deform under normal service conditions.

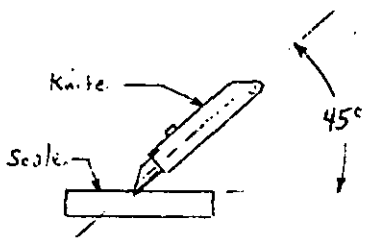


FIGURE 2. Blade and handle pressure test.

4.5.5 Adjustment guide test. With the blade locked first in the number 2 and then number 3 positions, hold the knife handle perpendicular to a scale surface and apply 50 lbs. (min.) of force downward on the tip of the blade for at least 5 seconds. During this test, the blade adjustment guide shall not release from its locked position, nor the blade break or slip out of the adjustment guide.

4.5.6 Blade brittleness test. With the knife handle mounted in a vise or similar clamping device and the blade fully extended, apply 20 lbs. of force at the tip of the blade in a direction perpendicular to the flat of the blade. During this test, the blade shall not break or take a permanent set.

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4.6 Inspection of preparation for delivery. Unless otherwise specified, inspection to determine compliance with preparation for delivery requirements shall be accomplished in accordance with PPP-P-40. Inspection level shall be level II with an AQL of 2.5 percent defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be level A, B, or C, as specified (see 6.1).

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following data information in procurement documents.

(a) Title, number, and date of this specification.
(b) A sheath (holster) if required (see 3.10).
(c) Level of packaging and packing required (see 5.1).

MILITARY INTERESTS:

DOD has waived coordination on revisions and amendments to this Federal Specification until further notice.

Preparing activity:

GSA-FSS

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